

**Module-05****RBT Level-L1, L2**

**Storytelling** - What is Storytelling, Characteristics of a Good Story, How to Create Data Stories, Components of Data Storytelling, Types of Data, What Makes a Good data story, Types of Charts for Data Visualization, Future of Data Storytelling

**Data storytelling** is the practice of combining data visualizations, narrative, and context to communicate insights clearly and drive action.

**Key Aspects of Data Storytelling**

- **Narrative (Context):** Explains the "why" behind the data, offering insights on what the data means and suggesting actions.
- **Visuals (Evidence):** Uses charts, graphs, and maps to highlight trends and patterns that support the narrative.
- **Data (Foundation):** Provides the underlying facts, analytics, and metrics that ensure credibility.

**Characteristics of a Good Story****❖ Define objectives.**

- Understand the purpose of data story, audience, and priorities.

**❖ Present a compelling narrative.**

- Know what audience wants to hear.

**❖ Incorporate key elements of analysis storytelling:** These elements include:

- **Plot:** The types of questions presented, how audience answer them, and how story will help audience arrive at the conclusion.
- **Context:** How audience interprets the data present.
- **Characters:** The tone and insights into how tell the story.
- **The end:** The conclusion and subsequent actions resulting from what the audience learned, how to achieve the results

they want, or what they'll need to do differently to improve for the future.

❖ **Be objective.**

- Present data transparently and objectively, avoid ambiguity, and make sure designs and visualizations don't compromise the integrity of the data or story.

❖ **Choose suitable types of visualizations.**

- It need to present information appropriately with the right imagery.

❖ **Follow graphic design best practices.**

- So audience completely understands, need to choose a presentation that reduces the most friction between reading and interpreting the graphical elements.

❖ **Use the proper data visualization tools.**

- To effectively communicate data through visualizations, need to use the right tool.

❖ **Make an insightful and human story.**

- If story is relatable, engaging, and offers high-value content in bite-sized pieces, keep audience's attention.

❖ **Create a story that complements both storytelling and visualizations.**

- Adding supporting, self-explanatory graphics to conceive a consistent storyline adds interest and increases engagement.

### **How to Create Data Stories?**

- ❖ **Know Objective:** Begin by clearly stating the goal of data story. What do want to convey or achieve with data? Is it to inform, persuade, or educate audience? Understanding your objective is crucial for crafting a compelling narrative
- ❖ **Know Audience:** Understand who audience is and what they are interested in. Tailor data story to their level of expertise, interests and needs. This will help you create a more relevant and engaging story

- ❖ **Data Gathering and preparation:** Collect and prepare the necessary data. Make sure that data is accurate, current and applicable to your goal. It may be necessary to clean, transform and analyze the data in order to derive insights from the dataset
- ❖ **Identify Key Insights:** Discover meaningful insights, trends, patterns, or anomalies within data. These valuable findings will form the foundation of your data narrative. Utilize data visualization tools and statistical analysis techniques to revealing these valuable insights
- ❖ **Set a narrative:** Structure the data story in a proper flow, with a clear beginning, middle and end. Begin by introducing the topic and setting the stage. Then, in the middle, present the main insights and findings. Finally, end with a summary and discuss the implications. Engage your audience by using storytelling techniques
- ❖ **Focus on data visualization:** Select the appropriate data visualization techniques to convey your insights efficiently. Use charts, graphs, maps and infographics to make your data visually appealing and easy to understand. Do ensure that visualization aligns with your narrative.
- ❖ **Share with Others:** Consider distributing data story through suitable channels like reports, presentations, websites or social media platforms. Take into account the most suitable format for audience, whether be it a PDF report, a live presentation or an interactive online dashboard

### Components of Data Storytelling

- ❖ **Data:** Data storytellers collect and preprocess the necessary data to narrate a story. They conduct statistical analysis and visually represent significant trends and patterns for comprehensive data examination.
- ❖ **Narrative:** Data storytellers collect and preprocess the data to narrate a story. They conduct statistical analysis and visually represent significant trends and patterns for comprehensive data examination.
- ❖ **Visuals:** A picture carries significant meaning. Adding visuals enhances the storytelling and makes the data more impactful. Visuals can include graphs, images, or videos.

## Types of Data

Data is mainly classified into two categories:

- ❖ Qualitative Data
- ❖ Quantitative Data

**Qualitative data** is descriptive, non-numerical information that focuses on the qualities, characteristics, and meanings of phenomena, rather than measurements or statistics.

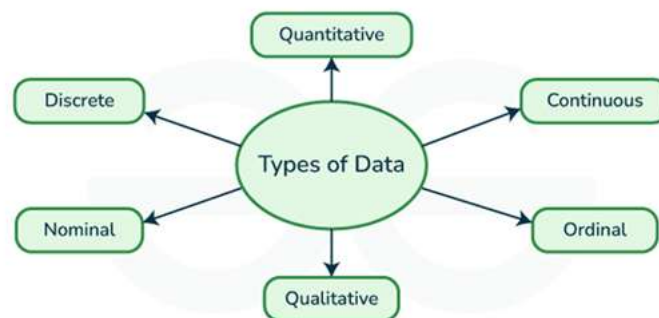
### Key Characteristics

- ❖ **Descriptive:** It describes attributes
- ❖ **Non-numerical:** Cannot be measured or counted directly.
- ❖ **Contextual:** Highly focused on context, experiences, and meaning.
- ❖ **Subjective:** Often relies on interpretation

**Quantitative data** is information that can be measured, counted, and expressed in numerical form, allowing for statistical analysis.

### Key Characteristics of Quantitative Data

- ❖ **Numerical:** Expressed in numbers (integers or decimals).
- ❖ **Objective:** Represents concrete, measurable facts rather than interpretations.
- ❖ **Analyzable:** Suitable for mathematical calculations and statistical tools.
- ❖ **Structured:** Often organized in spreadsheets, databases, or reports.



### Quantitative Data

- ❖ Discrete Data
- ❖ Continuous Data

### Qualitative Data

- ❖ Nominal Data
- ❖ Ordinal Data

**Discrete Data:** This type of data represents the information that can be counted and measured in a limited number of separate values. The values are usually whole numbers and come through counting/categorizing them. This type of data is actually a bit different from continuous data, which holds unlimited value within a specified range.

**Examples:** Number of students present in a class, Number of bikes present in a parking

**Continuous Data:** This type of data holds the information that can have any value within a specific range. Unlike the discrete data which has separate and distinct values, continuous data can be divided into smaller and more precise values.

**Examples:** Temperature, Time

**Nominal Data:** Nominal data is a form of categorical data that represents categories or labels without any inherent order or ranking. In nominal data, the categories are separate and do not have a specific order or numerical value assigned to them. This kind of data is qualitative in nature, and the categories are used to classify items or observations into groups.

**Examples:** Types of Car, Types of fruits

**Ordinal Data:** Ordinal data is a special kind of categorical data that shows categories with a natural order or ranking. Unlike nominal data, where

categories have no specific order, ordinal data allows for a meaningful comparison of values based on their relative position or rank.

**Examples:** Education Levels, Educational Grades

### What Makes a Good data story?

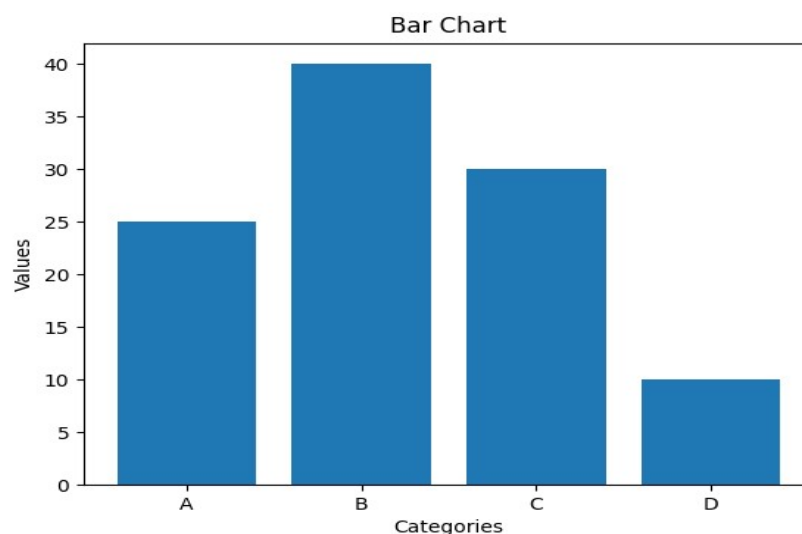
- ❖ **Clear Objective (Purpose):** A clear objective defines the main goal of the data story and what message needs to be communicated to the audience. It helps focus the analysis and ensures that only relevant data and visuals are used.
- ❖ **Audience Understanding:** Audience understanding means knowing who your audience is, their level of knowledge, and what they need from the data story. It helps in choosing the right language, level of detail, and type of visuals to communicate effectively
- ❖ **Relevant and Clean Data:** Relevant and clean data means using accurate, complete, and well-organized data that directly supports the objective of the story. It involves removing errors, duplicates, and unnecessary information to avoid confusion or misleading insights.
- ❖ **Effective Visuals:** Effective visuals present data in a clear, simple, and meaningful way that enhances understanding.
- ❖ **Strong Narrative (Story Flow):** A strong narrative provides a clear and logical flow to the data story, guiding the audience from beginning to end. It starts with context, moves through analysis, and ends with key insights or conclusions.
- ❖ **Highlight Key Insights:** Highlighting key insights means drawing attention to the most important findings in the data.
- ❖ **Simplicity and Clarity:** Simplicity and clarity mean presenting data in a straightforward and easy-to-understand manner without unnecessary complexity. It involves using clean visuals, clear labels, and concise explanations to avoid confusion.

- ❖ **Emotional and Contextual Connection:** Emotional and contextual connection means linking data to real-world situations and human experiences to make it more meaningful and engaging. Providing context helps the audience understand why the data matters, while an emotional connection makes the story more memorable.
- ❖ **Actionable Insights:** Actionable insights mean providing clear recommendations or decisions based on the data analysis.
- ❖ **Consistency and Design:** Consistency and design refer to maintaining a uniform and visually appealing layout throughout the data story. This includes using consistent colors, fonts, styles, and formatting to create a professional and cohesive look.

## Types of Charts for Data Visualization

### 1. Bar Charts

Bar charts are used to compare values across different categories using rectangular bars. X-axis shows categories while Y-axis represents values. Common types include horizontal, stacked and grouped bar charts.



### When to Use:

- To compare different categories

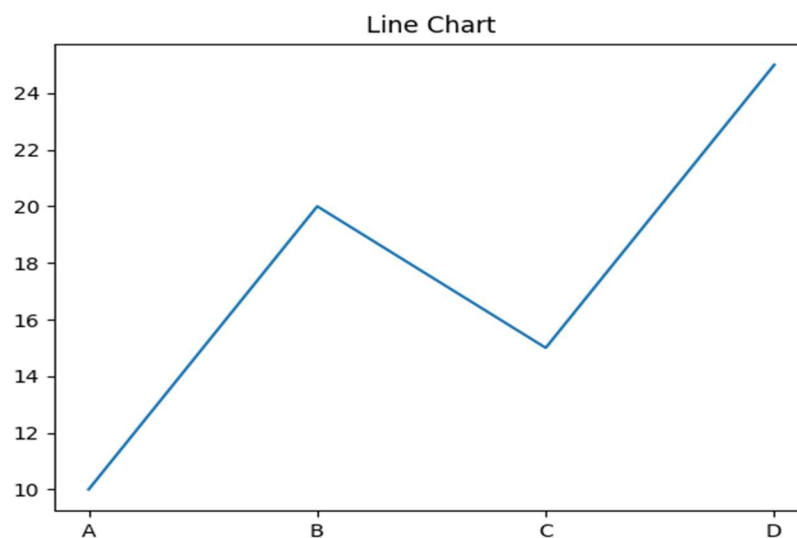


- To rank values from highest to lowest
- To show relationships between multiple variables

## 2. Line Charts

Line charts show how values change over time by connecting data points with lines. They help visualize trends like increases, decreases or stability.

Below is the example of line chart:



### When to Use:

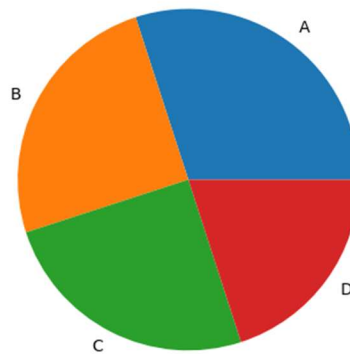
- To track changes over time
- To compare trends across multiple data series
- For time series analysis

## 3. Pie Charts

Pie charts are round charts divided into slices, where each slice shows a part of the whole. The size of each slice represents its percentage.

Below is the example of pie chart:



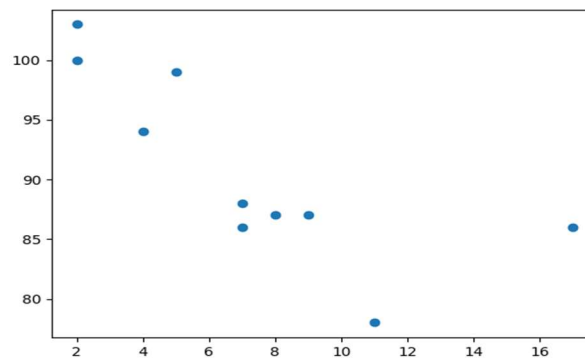
**When to Use:**

- To show how different parts contribute to a whole
- To highlight a dominant category

**4. Scatter Chart (Plots)**

Scatter charts use dots to show relationship between two numerical variables. X-axis shows the independent variable and Y-axis shows the dependent variable.

Below is the example of scatter chart:

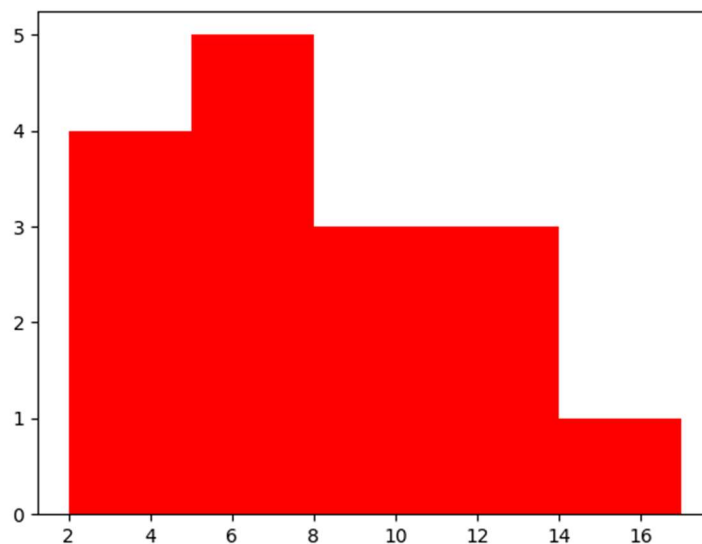
**When to Use:**

- To observe relationships between two variables
- To detect patterns, clusters or outliers in data

## 5. Histogram

A histogram displays the distribution of numerical data by grouping values into intervals (bins) and showing their frequency as bars. It helps reveal the shape, spread and patterns in the data.

Below is the example of histogram:



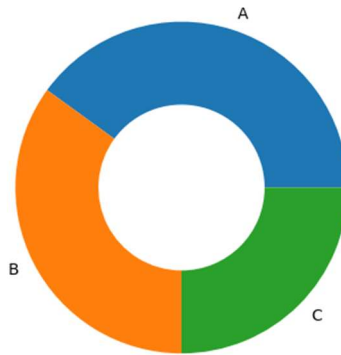
When to Use:

- To visualize the distribution of numerical data
- To explore patterns, trends and outliers

## 6. Donut or Doughnut chart

A donut chart is a circular chart like a pie chart but with a hole in the center. Each slice shows a category's contribution to the whole, making it visually cleaner and ideal for comparisons.

Below is the example of a donut chart:

**When to Use:**

- Show part-to-whole relationships like sales or market share.
- Track progress toward a goal (e.g., 70% completed).
- Best when comparing a few categories.

**Future of Data Storytelling**

- Data visualisation tools are evolving with the integration of augmented and virtual reality technologies, resulting in augmented data visualisation. This allows users to engage with data in immersive ways, making data storytelling more compelling and accessible.
- As AI and machine learning advance, they will become increasingly important in analyzing and interpreting data. This could result in automated data storytelling, where AI generates insights and narratives from raw data, enabling quicker and more data-focused decision-making.
- Users will be able to explore data stories in a nonlinear and personalized way as interactive storytelling tools become more advanced. This may include dynamic dashboards, interactive infographics, and adaptable storytelling that responds to user input and preferences.
- In light of increasing concerns regarding data privacy and ethics, data storytelling must tackle these matters. Future data storytellers must be cautious of data privacy regulations and ethical considerations while presenting data to the public.

| S.No | Questions                                                                                                      |
|------|----------------------------------------------------------------------------------------------------------------|
| 01   | <b>Define</b> Data storytelling? <b>List</b> and <b>explain</b> key Aspects & Characteristics of a Good Story. |
| 02   | <b>Explain</b> How to Create Data Stories?                                                                     |
| 03   | <b>List</b> and <b>explain</b> Components of Data Storytelling.                                                |
| 04   | <b>Explain</b> Types of Data in detail?                                                                        |
| 05   | <b>Explain</b> what Makes a Good data story?                                                                   |
| 06   | <b>Explain</b> Types of Charts for Data Visualization?                                                         |
| 07   | <b>Explain</b> Future of Data Storytelling.                                                                    |

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